

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q21-1872

Pinnacles 138 kV  
February 24, 2023

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1872. The Interconnection Customer proposes a 42.6 MW battery energy storage facility interconnecting at the Pinnacles 138 kV substation in the Virginia area. The request was received on August 29, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 562 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$264,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                 | Contingency                                           | Loading (%) | Rating (MVA) | DFAX (%) |
|------------------------------------|-------------------------------------------------------|-------------|--------------|----------|
| No Troy-Oconto 138.0 kV            | 4 Ac Sta-Ave 52 138.0 kV                              | 116.4       | 290          | 14.4     |
| Springbank-Oconto 138.0 kV L       | Ave 52-4 Ac Sta 138.0 kV line                         | 116.0       | 305          | 14.4     |
| Silver City-Fries 138.0 kV         | Sta 11-Stock Island 138.0 kV                          | 104.6       | 249          | 5.3      |
| Fracisco-Fries Substation 138.0 kV | Sta 11-Stock Island 138.0 kV                          | 104.4       | 250          | 5.3      |
| Sta 11-Springbank 138.0 kV         | Kean Mtn And Grassy Crk-Barr Lake 345.0 kV            | 101.8       | 483          | 8.2      |
| Stock Island-Station 11 138.0 kV   | Kean Mtn And Grassy Crk Substation-Barr Lake 345.0 kV | 101.4       | 504          | 8.2      |
| Francisco-Buck Hydro 138.0 kV Line | Sta 11 Substation-Stock Island 138.0 kV               | 100.9       | 247          | 5.3      |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------|---------------------------------------------|---------------|-----------------------|
| No Troy-Oconto 138.0 kV | Rebuild line with higher-capacity conductor | \$1,327,000   | 2% (\$23,000)         |

| Facility                                 | Required Upgrade                            | Cost Estimate | Allocation to Project |
|------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Springbank-Oconto 138.0 kV L             | Rebuild line with higher-capacity conductor | \$7,169,000   | 2% (\$125,000)        |
| Silver City-Fries 138.0 kV line          | Reconductor line                            | \$1,958,000   | 1% (\$21,000)         |
| Fracisco-Fries Substation 138.0 kV       | Reconductor line                            | \$888,000     | 1% (\$9,000)          |
| Springbank-Sta 11 138.0 kV               | Reconductor line                            | \$6,496,000   | 1% (\$40,000)         |
| Stock Island-Station 11 138.0 kV         | Reconductor line                            | \$1,423,000   | 1% (\$9,000)          |
| Francisco-Buck Hydro Substation 138.0 kV | Reconductor line                            | \$3,475,000   | 1% (\$37,000)         |
|                                          | Total allocated to project                  |               | \$264,000             |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.